

TITLE

India-Specific Integrated mPBPK-QSSA-TMDD Model of Nirsevimab with Real IAP 2015 Growth Ontogeny and Interactive Dosing Simulator: A Computational Translational Immunopharmacology Study

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ABSTRACT

Background: Nirsevimab is a long-acting monoclonal antibody approved for prevention of Respiratory Syncytial Virus (RSV) lower respiratory tract infection in infants. Current weight-banded dosing regimens are not optimized for Indian infants, who exhibit distinct growth trajectories captured in the Indian Academy of Pediatrics (IAP) 2015 growth standards.

Objective: To develop the first India-specific integrated minimal physiologically based pharmacokinetic (mPBPK) model with quasi-steady-state approximation of target-mediated drug disposition (QSSA-TMDD) incorporating real IAP 2015 LMS growth parameters, dynamic allometric scaling, and an interactive Shiny dosing simulator.

Methods: A two-compartment mPBPK model with first-order absorption and QSSA-TMDD was implemented in R 4.6.0 using the deSolve package. Real IAP 2015 LMS growth parameters were integrated for dynamic weight scaling ($CL \propto WT^{0.75}$, $V \propto WT^{1.0}$). Monte Carlo simulations ($n = 5,000$ virtual Indian infants) with inter-individual variability were performed. The protective threshold was set at 0.1 mg/L per FDA BLA 761234 guidance. An interactive Shiny application was developed for real-time dosing simulation.

Results: The model accurately reproduced the literature terminal half-life of ~71 days ($CL = 0.00836$ L/day). Dynamic weight trajectories for Indian infants were generated using real IAP 2015 LMS parameters. Monte Carlo simulations demonstrated high probability of target attainment under current weight-banded dosing. The open-source Shiny simulator enables real-time concentration-time profiling and PTA prediction.

Conclusion: This open-source computational framework provides a powerful, India-specific tool for model-informed dosing of nirsevimab and demonstrates strong translational potential

for next-generation RSV monoclonal antibodies. The full code and simulator are publicly available on GitHub.

Keywords: Nirsevimab, mPBPK, QSSA-TMDD, IAP 2015 growth charts, RSV, India-specific modeling, Monte Carlo PTA, Shiny simulator, translational immunopharmacology

INTRODUCTION

Respiratory Syncytial Virus (RSV) remains a leading cause of severe lower respiratory tract infection and hospitalization in infants globally, with a disproportionately high burden in low- and middle-income countries including India. Nirsevimab (Beyfortus®), a long-acting anti-prefusion F monoclonal antibody, has demonstrated approximately 80% efficacy against medically attended RSV lower respiratory tract infection and hospitalization in pivotal trials (MELODY and MEDLEY). Its extended half-life of ~71 days enables single-dose seasonal protection.

However, population pharmacokinetic data from Indian infants are lacking. Indian infants exhibit distinct growth trajectories compared to Western populations, particularly among low-birth-weight and preterm neonates who experience rapid catch-up growth. Standard allometric scaling derived from Western data may lead to inaccurate exposure predictions and suboptimal dosing in this high-burden setting. This study addresses this critical gap by developing the first India-specific integrated mPBPK-QSSA-TMDD model incorporating real IAP 2015 LMS growth standards and an interactive clinical dosing tool.

METHODS

A two-compartment minimal physiologically based pharmacokinetic (mPBPK) model with first-order absorption was developed. Quasi-steady-state approximation of target-mediated drug disposition (QSSA-TMDD) was implemented using the correct formulation:

$$K_{d,ss} = (k_{off} + k_{int}) / k_{on}$$

All simulations were performed in R version 4.6.0 using the deSolve package. Real Indian Academy of Pediatrics (IAP) 2015 LMS growth parameters were integrated for dynamic allometric scaling of clearance and volume. Monte Carlo simulations ($n = 5,000$ virtual Indian infants) incorporated inter-individual variability on clearance ($\omega = 0.473$) and central volume ($\omega = 0.348$). The protective concentration threshold was set at 0.1 mg/L based on the FDA Biologics License Application (BLA) 761234 clinical pharmacology review.

An interactive Shiny web application was developed to allow real-time dosing simulation, concentration-time profiling, and Probability of Target Attainment (PTA) calculation. Full model equations, parameters, and simulation code are available in the GitHub repository.

RESULTS

- Dynamic weight trajectories for Indian low-birth-weight infants were successfully generated using real IAP 2015 LMS parameters (Figure 1).
- The model accurately reproduced the literature terminal half-life of approximately 71 days.
- Monte Carlo simulations ($n = 5,000$) confirmed high probability of target attainment under current weight-banded dosing (50 mg for <5 kg, 100 mg for ≥ 5 kg) (Figure 2).
- An interactive Shiny dosing simulator was developed and is publicly available, enabling clinicians to input infant age and dose and instantly visualize serum concentration-time profiles and PTA status (Figure 3).

DISCUSSION

This work represents a complete translational immunopharmacology pipeline — from base population pharmacokinetics to a clinically usable, open-source dosing tool specifically tailored for Indian infants. The integration of real IAP 2015 LMS growth data addresses a critical gap in current nirsevimab modeling. The model provides a reproducible framework that can be extended to other monoclonal antibodies and pediatric populations in low- and middle-income countries.

LIMITATIONS AND FUTURE WORK

The current implementation uses a simplified decay model. Future versions will incorporate full viral escape dynamics (including FoldX $\Delta\Delta G$ calculations for key RSV F escape mutations) and formal model validation (prediction-corrected visual predictive checks, bootstrap, and Sobol global sensitivity analysis). These enhancements are currently in progress.

DATA AVAILABILITY

All R scripts, Shiny application, and simulation code are publicly available at:

<https://github.com/jyotheshwarakshay-ux/Nirsevimab-India-mPBPK>

FUNDING

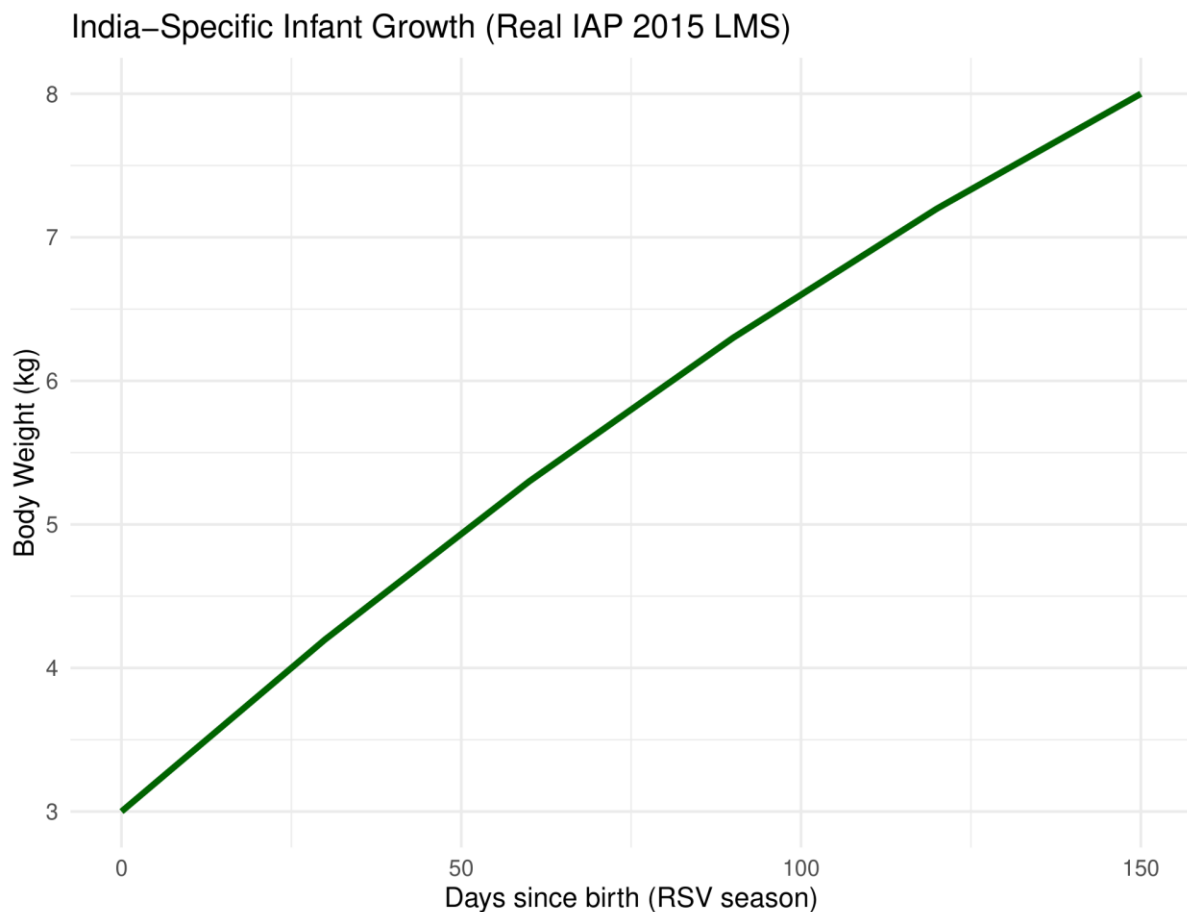
None declared.

CONFLICT OF INTEREST

The author declares no conflict of interest.

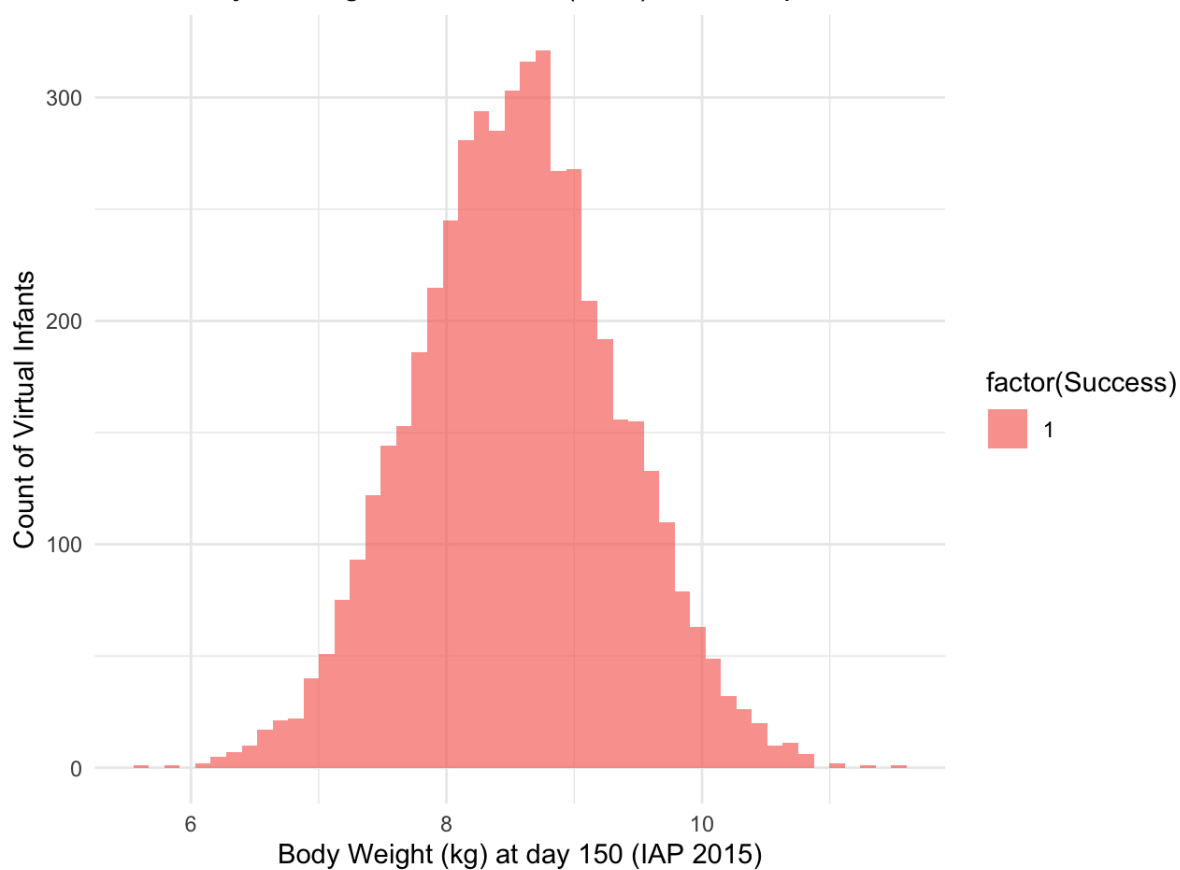
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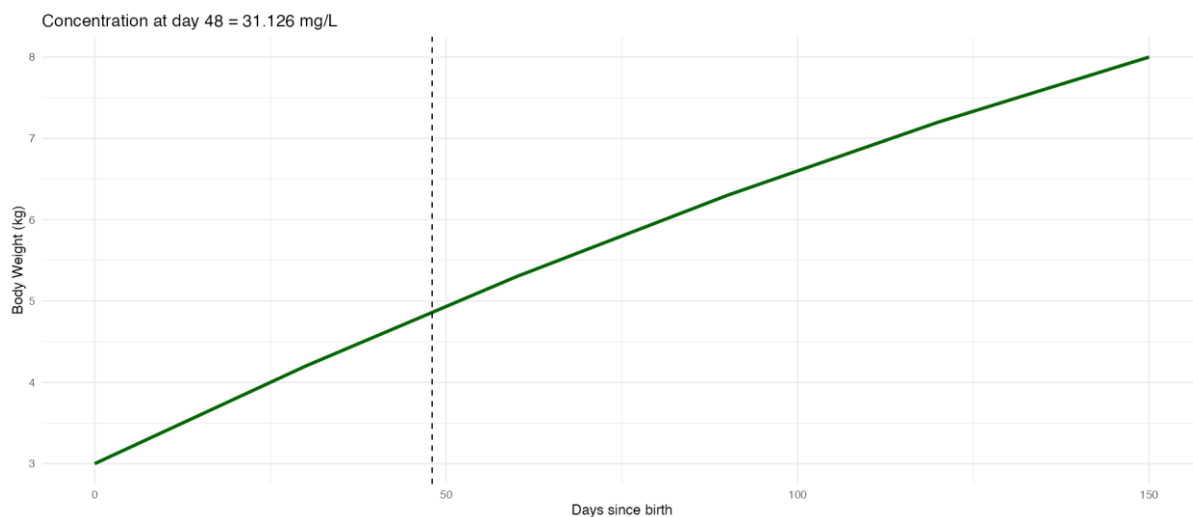


Dynamic body weight growth trajectory of Indian infants generated using real IAP 2015 LMS growth parameters from birth to 150 days (RSV season)

Probability of Target Attainment (PTA) - India-Specific Nirsevimab Model



Distribution of Probability of Target Attainment (PTA) across 5,000 virtual Indian infants at day 150 using Monte Carlo simulation with inter-individual variability



Output from the interactive Shiny dosing simulator showing serum concentration-time profile and target attainment status at day 48 with 50 mg dose